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Skill week – 6

main.py code:

```
from fastapi import FastAPI, HTTPException, Depends
```

```
from fastapi.security import (
```

```
OAuth2PasswordBearer,
```

```
OAuth2PasswordRequestForm )
```

```
from jose import jwt
```

```
app = FastAPI()
```

```
SECRET_KEY = "my-secret-key"
```

```
ALGORITHM = "HS256"
```

```
oauth2_scheme = OAuth2PasswordBearer(
```

```
tokenUrl="login"
```

```
)
```

```
@app.post("/login")
```

```
def login(
```

```
form_data: OAuth2PasswordRequestForm = Depends()
```

```
):
```

```
if (
```

```
form_data.username == "Sujit"
```

```
and form_data.password == "Sujit@2026"
```

```
):  
payload = {  
  
    "sub":  
  
    form_data.username,  
  
    "role": "student"  
  
}  
  
token = jwt.encode(  
  
    payload, SECRET_KEY,  
  
    algorithm=ALGORITHM  
  
)  
  
return {  
    "access_token": token,  
  
    "token_type": "bearer"  
  
}  
  
raise HTTPException(  
  
    status_code=401,  
  
    detail="Invalid username or password"  
  
)  
  
def get_current_user(  
  
    token: str = Depends(oauth2_scheme)  
  
):
```

```
try:

    payload = jwt.decode(

        token,

        SECRET_KEY,

        algorithms=[ALGORITHM]

    )

    return payload

except Exception:

    raise

    HTTPException(

        status_code=401,

        detail="Invalid or expired token"

    )

@app.get("/profile")

def profile(

    user=Depends(get_current_user)

):

    return {

        "message": "Welcome to your profile",

        "user_id": user["sub"],
```

```
"role": user["role"]
```

```
}
```

POST

POST /login Login

Parameters

No parameters

Request body *required*

grant_type
string | (string | null)
pattern: ^password\$

☐ Send empty value

username * *required*
string

password * *required*
string(\$password)

☐ Send empty value

scope
string

☒ Send empty value

client_id
string | (string | null)

☐ Send empty value

client_secret
string | (string | null)(\$password)

☐ Send empty value

Execute

Response –

```
Curl
curl -X 'POST' \
  'http://127.0.0.1:8000/login' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/x-www-form-urlencoded' \
  -d 'grant_type=password&username=Charan123&password=priyanka123&scope=&client_id=string&client_secret=*****'

Request URL
http://127.0.0.1:8000/login

Server response
Code    Details
200
Response body
{
  "access_token": "eyJhOGciOiJ1IjE1N1IsInR5cCI6IkpXVCJ9.eyJzdWIiOiQmdSVM6iY51sInZv6Uj01ZkdV26C8IiwgcGl0aWV2Oj04VncX_x-7k2c1D0f8m8qk1DnJ00knc",
  "token_type": "bearer"
}
Response headers
content-length: 169
content-type: application/json
date: Tue, 15 Sep 2026 05:35:18 GMT
server: uvicorn
```

Authorize

Available authorizations ✕

Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes.
API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Authorized

Token URL: login
Flow: password
username: Sujit
password: *****
Client credentials location: basic
client_secret: *****

LogoutClose

Get :

GET

/profile Profile

Parameters

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8080/profile' \
-H 'accept: application/json' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIjOiJkcml5VSI6InJvbGUiOiJzdmR1bn60nd. sgVioqQwDwV1o4VbvX_x_-7k2c1D0FBm8qk1Dn200knc'
```

Request URL

```
http://127.0.0.1:8080/profile
```

Server response

Code

Details

200

Response body

```
{
  "message": "Welcome to your profile",
  "user_id": "Sujit",
  "role": "student"
}
```

Download

Response headers

```
content-length: 75
content-type: application/json
date: Tue, 15 Sep 2026 05:40:14 GMT
server: uvicorn
```